FIND Command with Examples:

Searching a file with a specific name under PWD

```
#find    .        -name        <F/N>
                              Pragathi
```

Searching files under Home dir

```
#find    /home    -name        <FN>
                               File1
```

Searching files with specific name & Ignoring Case

```
#find    /home        -iname       <FN>
                                    File2
```

Searching directories with specific name

```
#find    /        -type    d        -name      <dirname>
                                                Oracle
```

Searching files with specific PERM

```
#find    .        -type    f        -perm   644
```

Searching the files with out 777

```
#find    /        -type    f        !        -perm    777
```

Searching SUID files

```
#find    /        -perm   /u=s
```

Searching SGID files

```
#find   /   -perm /g=s
```

Searching empty files under specific dir

```
#find    /tmp          --type   f        --empty
```

Searching empty directories

```
#find    /tmp          -type    d        --empty
```

Searching hidden files

```
#find          /tmp          -type    f        -name   ".*"
```

Seaching all files based on user

```
#find    /home  -user    steve
```

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Find all files based on group
#find /home -group g5

Searching modified files in last one hour
#find / -mmin -60

Searching accessed files in last one hour
#find / -amin -60

(Displays inode number of specific file)
#ls -i <FN>
Abc1

O/P
6568 abc1

#ls -R /sun6
(Displays FS available Inodes Information)

#df -e

Inode Number:

It is the unique number assigned to every file. For every file it has file name and inode number.File name is for user reference and inode number is for kernel reference.
NOTE: Inode number is unique for every file.

NOTE : Inode number contains every information about the files like :

Owner of the file
Group owner of the file
Perm of the file
Number of links
Size of the file

#ls -i <FN>
Abc1

O/P
6568 abc1

LINKS:

Links are pointers

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Connection between two strings

2 types of Links

1)Hard Links
2)Soft Links

Hard Links:
The two files having hard links will have the same inode number. If we update the data to the source file the same data we can access from the destination file & vice versa.
If we delete the source file we can access the data from the destination file with the same inode number. Hard links works as duplicate copies.

Limitations:
1)We cannot identify the files having hard links
2)hard links are applicable for only files
3)Hard links need to be configured with in the same filesystem.

#ln <SF> <DF>
Cap11 cap12

Soft Links :

The two files having soft link will be having different inode number. If we update the data to the source file the same data we can access from the destination file & vice versa.
If we delete the source file we cannot access the data from the destination file.Soft links acts as short cuts.

Advantages:
1)We can identify the files having soft links by seeing its perm.
2)Soft links are applicable for both files & directories.
3)Soft links can span across multiple filesystems.

#ln -s <SF> <DF>
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Paths:

We have 2 different types of paths:
1)Absolute Path : The path which is reference to the root(/) directory

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#mkdir –p /sap/data/db

#cd /sap/data/db

#pwd
O/P
/sap/data/db

2) Relative Path : The path which is reference to the home or current working directory other than root.

#mkdir -p /sap/test/objects

#pwd
O/P
/sap/data/db

#cd ../../test/objects

#pwd
O/p
/sap/test/objects

File Archiving :

The Linux "tar" stands for tape archive, which is used by large number of Linux/Unix system administrators to deal with tape drives backup. The tar command used to rip a collection of files and directories into highly compressed archive file commonly called tarball or tar, gzip and bzip in Linux. The tar is most widely used command to create compressed archive files and that can be moved easily from one disk to another disk or machine to machine.

1)Tar
2)gzip
3)gz2
4)xz

TAR

Creating an archive named archive.tar from file1,file2,file3
#tar -cf archive.tar file1 file2 file3

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Destination source

-c Create an archive
-f --file name, Specifies the destination file to archive

Listing the contents for archive:

#tar -tf archive.tar

Extracting files from an archive:

#mkdir /test
#cd /test
#tar -xf archive.tar

Create gzip compressed archive named /root/etc.tar with the /etc directory content :

#tar -cf /root/etc.tar /etc

#tar -tf /root/etc.tar (Listing)

Extracting :

#mkdir /root/etcbackup

#cd /root/etcbackup

#tar -xf /root/etc.tar

Create gzip compressed archive named /root/etcbackup.tar.gz with contents of /etc directory:

Gzip

#tar -czf /root/etcbackup.tar.gz /etc

-z -Compressing in gzip format

#tar -tf /root/etcbackup.tar.gz (Listing)

#mkdir /etcrestore (Extracting)
#cd /etcrestore
#tar -xzf /root/etcbackup.tar.gz

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BZ2
Create Bzip2 compressed archive named /root/logbackup.tar.bz2 with the contents of /var/log directory:

#tar -cjf /root/logbackup.tar.bz2 /var/log

-j ---Compressing in bz2 format

#tar -tf /root/logbackup.tar.bz2 (Listing)

#mkdir /logrestore (Extract)

#cd /logrestore
#tar -xjf /root/logbackup.tar.bz2

XZ
Create .xz compressed archive named /root/man.tar.xz from the contents of /usr directory:

#tar -cjf /root/man.tar.xz /usr

#tar -tf /root/man.tar.xz (Listing)

#mkdir /manrestore (extract)

#cd /manrestore

#tar -xjf /root/man.tar.xz

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